

MEWS Development Program

Software Requirements Specification (SRS) — Sample Portfolio Artifact

1. Purpose

This sample SRS defines the software requirements for an integrated medical early-warning and patient-monitoring platform. It describes the expected software behavior, interfaces, data handling, quality considerations, and acceptance expectations used to guide design, implementation, integration, verification, and validation.

2. System Scope

The software scope covers the mobile application, backend services, cloud platform, monitoring workflows, physiological-data processing, device/gateway connectivity, and supporting interfaces required for the integrated monitoring solution.

3. System Context

Wearable Device → Gateway / Connectivity → Cloud Services → Backend / Data Services → Mobile / Monitoring Application

The software operates as part of a larger hardware-and-software system. Requirements therefore include dependencies on device behavior, connectivity, cloud services, external technology partners, quality controls, and validation activities.

4. Functional Requirements

ID	Requirement
FR-01	The system shall receive physiological monitoring data from supported connected devices through the defined connectivity path.
FR-02	The system shall associate received observations with the appropriate patient / encounter context.
FR-03	The system shall process supported physiological observations for monitoring and early-warning-score workflows.
FR-04	The application shall present current monitoring information to authorized users.
FR-05	The system shall retain appropriate historical observation information for supported monitoring workflows.
FR-06	The system shall support device, gateway, and configuration information required for operation.
FR-07	The system shall provide application interfaces required by connected platform components.
FR-08	The system shall support error, event, and operational logging appropriate to system operation.
FR-09	The system shall protect sensitive healthcare information using appropriate authentication, authorization, and secure data-transfer mechanisms.
FR-10	The system shall support integration and validation activities through testable and traceable requirements.

5. Non-Functional Requirements

Area	Expectation
Security	Sensitive data and interfaces shall be protected through appropriate authentication, authorization, encryption, and secure communications.
Performance	The system shall process and present monitoring information within performance expectations defined for the intended workflow.
Availability	Cloud and application components shall support the operational availability required by the monitoring workflow.
Traceability	Requirements shall be traceable to design, implementation, verification, validation, and applicable change records.
Maintainability	Software components shall be documented and structured to support controlled maintenance and future changes.
Auditability	Relevant system events and changes shall provide sufficient records to support operational and quality review.

6. Verification & Validation

Each approved requirement should be linked to objective verification or validation evidence. Depending on the requirement, evidence may include software testing, integration testing, device testing, usability or workflow validation, inspection, review, or other documented verification activities.

7. Requirements Traceability

Business / User Need → System Requirement → Software Requirement → Design Input → Implementation → Verification / Validation Evidence → Release Decision

8. Change Control

Requirement changes should be assessed for impact on scope, architecture, implementation, verification, validation, quality, regulatory, and downstream release activities before approval and implementation.

9. Portfolio Evidence

The following supplied product and prototype images are referenced as visual evidence for the physical system context represented by the requirements.

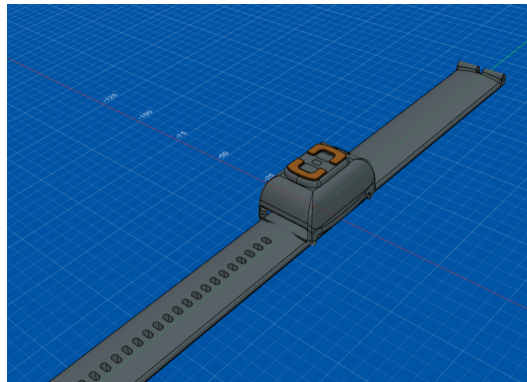


Figure 1 — Wearable device design reference

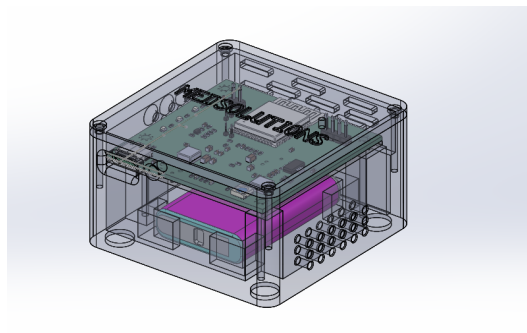


Figure 2 — Gateway enclosure design reference